

Solution first chapter the Standard Model and Beyond

Abstract

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1 Useful equations

A list of useful results

$$M = \frac{1}{2}\text{Tr}(M)\mathbb{1} + \frac{1}{2}\text{Tr}(M\sigma_i)\sigma_i$$

2 Exercise 1

Let $x_n, n = 1, \dots, 4$ be arbitrary Pauli spinors (i.e., two component complex column vectors). Then the bilinear form $x_m^\dagger \sigma_i x_n$ is an ordinary number for any choice of i, m, n . Prove the Fierz identity

$$(\chi_4^\dagger \vec{\sigma} \chi_3) \cdot (\chi_2^\dagger \vec{\sigma} \chi_1) = 2\eta_F(\chi_4^\dagger \chi_1)(\chi_2^\dagger \chi_3) - (\chi_4^\dagger \chi_3)(\chi_2^\dagger \chi_1)$$

To solve this, we make use of the identity (write identity in 'useful equations' sections) First of all, write

$$(\chi_4^\dagger \vec{\sigma} \chi_3)(\chi_2^\dagger \vec{\sigma} \chi_1) = \chi_4^\dagger \vec{\sigma} \chi_3 \chi_2^\dagger \vec{\sigma} \chi_1$$

Focusing on the middle, rewrite the $\chi_3 \chi_2^\dagger$ using the identity (mention)

$$\chi_3 \chi_2^\dagger = \frac{1}{2} \text{Tr}(\chi_3 \chi_2^\dagger) \mathbb{1} + \frac{1}{2} \text{Tr}(\chi_3 \chi_2^\dagger \vec{\sigma}) \vec{\sigma}$$

But, $\text{Tr}(\chi_3 \chi_2^\dagger) = \chi_2^\dagger \chi_3$, therefore

$$\begin{aligned} \chi_4^\dagger \vec{\sigma} \chi_3 \chi_2^\dagger \vec{\sigma} \chi_1 &= \chi_4^\dagger \vec{\sigma} \frac{1}{2} (\chi_2^\dagger \chi_3) \vec{\sigma} \chi_1 + \chi_4^\dagger \vec{\sigma} \frac{1}{2} \text{Tr}(\chi_3 \chi_2^\dagger \vec{\sigma}) \vec{\sigma} \chi_1 \\ &= \frac{1}{2} \chi_4^\dagger \chi_1 \chi_2^\dagger \chi_3 + \frac{1}{2} \chi_4^\dagger \vec{\sigma} \chi_1 \text{Tr}(\chi_3 \chi_2^\dagger \vec{\sigma}) \end{aligned}$$

The unknown part is solving for the trace of $\chi_3 \chi_2^\dagger \vec{\sigma}$. Let's take a closer look to this element. By components, we have

$$\left(\chi_3 \chi_2^\dagger \right)_b^a \vec{\sigma}_c^b = \left(\chi_2^\dagger \right)_b \chi_3^a \vec{\sigma}_c^b = \left(\chi_2^\dagger \right)_b \vec{\sigma}_c^b \chi_3^a = \left(\chi_2^\dagger \vec{\sigma} \chi_3 \right)_c^a$$

Therefore, we get that

$$\text{Tr} \left(\chi_3 \chi_2^\dagger \vec{\sigma} \right) = \chi_2^\dagger \vec{\sigma} \chi_3$$

With this, we rewrite the above expression as

$$2\chi_4^\dagger \vec{\sigma} \chi_3 \chi_2^\dagger \vec{\sigma} \chi_1 = \chi_4^\dagger \chi_1 \chi_2^\dagger \chi_3 + \chi_4^\dagger \vec{\sigma} \chi_1 \chi_2^\dagger \vec{\sigma} \chi_3$$

Now we just apply the same steps from above to $\chi_4^\dagger \vec{\sigma} \chi_1 \chi_2^\dagger \vec{\sigma} \chi_3$, obtaining

$$2\chi_4^\dagger \vec{\sigma} \chi_3 \chi_2^\dagger \vec{\sigma} \chi_1 = \chi_4^\dagger \chi_1 \chi_2^\dagger \chi_3 + \frac{1}{2} \chi_4^\dagger \chi_3 \chi_2^\dagger \chi_1 + \frac{1}{2} \chi_4^\dagger \vec{\sigma} \chi_3 \chi_2^\dagger \vec{\sigma} \chi_1$$

Nope, i get stuck ahahh

The second way is by expanding $\chi_1 \chi_2^\dagger$ in the middle of

$$\chi_4^\dagger \chi_1 \chi_2^\dagger \chi_3$$

With this, we obtain

$$\begin{aligned}
\chi_4^\dagger \chi_1 \chi_2^\dagger \chi_3 &= \chi_4^\dagger \chi_3 \chi_2^\dagger \chi_1 + \chi_4^\dagger \vec{\sigma} \chi_3 \text{Tr}(\chi_1 \chi_2^\dagger \vec{\sigma}) \\
&= (\chi_4^\dagger \chi_3)(\chi_2^\dagger \chi_1) + (\chi_4^\dagger \vec{\sigma} \chi_3)(\chi_2^\dagger \vec{\sigma} \chi_1)
\end{aligned}$$

which implies that

$$\left(\chi_4^\dagger \vec{\sigma} \chi_3 \right) \left(\chi_2^\dagger \vec{\sigma} \chi_1 \right) = 2 \left(\chi_4^\dagger \chi_1 \right) \left(\chi_2^\dagger \chi_3 \right) - \left(\chi_4^\dagger \chi_3 \right) \left(\chi_2^\dagger \chi_1 \right)$$

3 Exercise 2

Justify the fact that $\delta(f(x)) = \sum_i \frac{\delta(x-x_0^i)}{\left|\frac{df}{dx}\right|_{x_0^i}}$, where $f(x_0^i) = 0, \forall i$. Justify the fact that $\delta(f(x)) = \sum_i \frac{\delta(x-x_0^i)}{\left|\frac{df}{dx}\right|_{x_0^i}}$, where $f(x_0^i) = 0, \forall i$.

To solve this, we use the substitution techniques for integrals. So, we integrate

$$\int \delta(f(x)) \, dx$$

with the substitution

$$\begin{cases} y = f(x) \\ \frac{dy}{dx} = f'(x) \end{cases} \implies dy = f'(x) \, dx \implies \frac{1}{y'} dy = dx \quad .$$

We get

$$\int \delta(f(x)) \, dx = \int \delta(y) \frac{1}{y'} \, dy = \sum_{x|y(x)=0} \frac{1}{y'(x)} = \sum_i \int \frac{\delta(x-x_0^i)}{\frac{df}{dx}} \, dx$$

The δ is symmetric under $f(x) \rightarrow -f(x)$, so we should get the same result in either case, therefore

$$\delta(f(x)) = \sum_i \frac{\delta(x-x_0^i)}{\left|\frac{df}{dx}\right|_{x_0^i}}$$

as required

4 Exercise 3

Calculate the surface area $\int d\Omega_n$ of a unit sphere in a n dimensional euclidean space

The trick consists in using the gaussian exponential. First of all, an n -sphere is an object which is angle-symmetric, so we may as well suppose that

$$V_n = C_n R^n$$

that is, the volume is proportional to the radius through a constant which depends only upon the dimensionality¹. But, the volume element is $dV_n = S_n(R)dR$. Now, we do n gaussian integral, and see the magic:

$$\pi^{\frac{n}{2}} = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} e^{-x_k^2} \prod_k dx_k = \int_0^{+\infty} e^{-R^2} n C_n R^{n-1} dR = n C_n \frac{1}{2} \Gamma\left(\frac{n}{2}\right) = \left(\frac{n}{2}\right)! C_n$$

where we have identified $\prod_k dx_k = dV_n$. And that's it! We even get a usefule byproduct, which is how the surface of a sphere scales with dimension

$$V_n(R) = \frac{\pi^{\frac{n}{2}}}{\frac{n}{2}!} R^n$$
$$S_n(R) = \frac{2\pi^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} R^{n-1}$$

¹For these argument, and basically all the ideas, see e.g. Rogel-Salazar 2011

5 Exercise 4

Show that the generic Lorentz boost

$$E' = \gamma_L (E + \vec{\beta}_L \cdot \vec{p})$$

$$\vec{p}' = \vec{p}_L + \gamma_L (\vec{p}_\parallel + \vec{\beta}_L E)$$

can be written as

$$\begin{pmatrix} E' \\ p'_\parallel \end{pmatrix} = \begin{pmatrix} \cosh y_L & \sinh y_L \\ \sinh y_L & \cosh y_L \end{pmatrix} \begin{pmatrix} E \\ p_\parallel \end{pmatrix}$$

where $y_L = \frac{1}{2} \ln \frac{1+\beta_L}{1-\beta_L}$ is the rapidity of the boost

To solve this exercise, first we inspect what are $\sinh y_L$ and $\cosh y_L$. Substituting the expression of y_L , one finds that

$$2 \sinh y_L = e^{y_L} + e^{-y_L} = e^{\ln \sqrt{\frac{1+\beta_L}{1-\beta_L}}} e^{-\ln \sqrt{\frac{1+\beta_L}{1-\beta_L}}} + e^{-\ln \sqrt{\frac{1+\beta_L}{1-\beta_L}}} e^{\ln \sqrt{\frac{1+\beta_L}{1-\beta_L}}} = \sqrt{\frac{1+\beta_L}{1-\beta_L}} + \sqrt{\frac{1-\beta_L}{1+\beta_L}}$$

$$2 \cosh y_L = e^{y_L} - e^{-y_L} = \sqrt{\frac{1+\beta_L}{1-\beta_L}} - \sqrt{\frac{1-\beta_L}{1+\beta_L}}$$

Of course, this can't be the end of the story, so we take a closer look to the expression. A good tip with "similar looking" addition is to factorise the first member. In this case, the trick does its job since

$$\begin{aligned} \sqrt{\frac{1+\beta_L}{1-\beta_L}} + \sqrt{\frac{1-\beta_L}{1+\beta_L}} &= \sqrt{\frac{1+\beta_L}{1-\beta_L}} \overbrace{\left(1 + \frac{1-\beta_L}{1+\beta_L}\right)}^{\frac{2}{1+\beta_L}} \\ \frac{2}{\sqrt{(1+\beta_L)(1-\beta_L)}} &= \frac{2}{\sqrt{1-\beta_L^2}} = 2\gamma \sqrt{\frac{1+\beta_L}{1-\beta_L}} - \sqrt{\frac{1-\beta_L}{1+\beta_L}} = \sqrt{\frac{1+\beta_L}{1-\beta_L}} \left(\frac{2\beta_L}{1+\beta_L}\right) \\ &= 2\beta_L \gamma \end{aligned}$$

Therefore, we find that

$$\cosh y_L = \gamma$$

$$\sinh y_L = \beta_L \gamma$$

With this, we solve the system, which yields

$$\begin{cases} E' = \gamma(E + \beta_L p_\parallel) \\ p'_\parallel = \gamma(\beta_L E + p_\parallel) \end{cases}$$

Of course, this is unsatisfactory, since what we are trying to demonstrate is a relation between 4-vectors and not 2-vectors. Now, how do we retrieve the original identification? The trick is to identify²

$$\begin{cases} p_{\parallel} \rightarrow \vec{p} = \left(p_{\parallel} & \frac{1}{\gamma} \vec{p}_{\perp} \right) \\ \beta_L \rightarrow \vec{\beta}_L \end{cases}$$

and, finally, we see that

$$\begin{cases} E' &= \gamma(E + \vec{\beta}_L \cdot \vec{p}) \\ \vec{p}' &= \vec{p}_{\perp} + \gamma_L(\vec{p}_{\parallel}) \end{cases}$$

²I'm currently not aware if this is unique, since we are mapping a scalar to a vector. Of course we would have an $\mathbb{S}^1$ uncertainty which we can reduce to a $\mathbb{Z}^2$ uncertainty since we may as well specify the direction of the boost.

References

- [1] J Rogel-Salazar. *Statistical Mechanics, 3rd edn., by RK Pathria and PD Beale: Scope: textbook. Level: postgraduate or advanced undergraduate.* 2011.